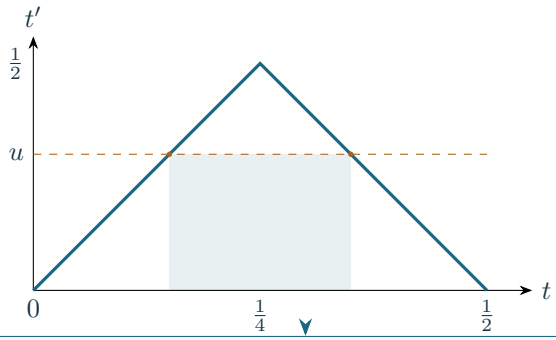
A proved local bound and its global use

Exact coefficient Cauchy inequality

q_k is a nonzero tail polynomial with nonnegative coefficients.
For every real λ and positive reference λ_0,

$$\frac{|q_k(\lambda)|^2}{q_k(\lambda_0)} \quad \leq \quad q_k(\lambda^2/\lambda_0).$$

The folded angle has a uniform distribution



Shown: $\delta = \frac{1}{4}$.

$t' = \frac{1}{\pi} \arctan |R_\delta(t)|.$

$R_\delta(t) = \frac{\sin(2\pi t)}{\cos(2\pi t) + \cos(2\pi\delta)}.$

For every δ,
$|\{t : t' > u\}| = \frac{1}{2} - u.$

Proved local estimate, at both level parities

$$\frac{1}{N} \log \frac{\sup_{\sigma \in [-2,2]} |a_N(v;\sigma)|^2}{p_N(v)} \quad \leq \quad C(q_x, q_y, \tfrac{1}{2} - q_x - q_y) + o(1).$$

$q = v/(N-1)$ is the original scaled degree; a_N is its coefficient,
p_N the positive endpoint reference, and C the triangle entropy.
The bound retains all original degree constraints.

A sufficient test for another diagram

Combine the local bounds while retaining every shared face load:

$$\max_{q \in \mathcal{P}_D^{\mathrm{NC}}} \sum_e \min\left\{ h_{\mathrm{env},e}(q_e), \frac{h_e(q_e) + g_e(q_e)}{2} \right\} \quad < \quad \Phi_D.$$

The minimum selects between two local upper rates.
$\mathcal{P}_D^{\mathrm{NC}}$ retains the shared degree constraints;
Φ_D is the complete action of diagram D.
Some matched-sphere diagrams fail this sufficient test.

Volume equality is already proved in specified classes

Strict Conway trees and the stated certified plat families satisfy
their volume theorems under their own long-twist hypotheses.

Those proofs do not assert that the displayed sufficient test
holds for every covered diagram.